

CHAPLEAU, ON, Canada

WMO: 716420

Lat: 47.820N

Lon: 83.347W

Elev: 1465

StdP: 13.93

Time Zone: -5.00 (NAE)

Period: 96-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.5	-20.1	-32.4	0.9	-24.1	-25.9	1.4	-17.2	19.0	20.9	17.1	18.7	2.0	320	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	82.8	67.7	80.3	66.2	77.0	63.8	71.2	80.1	69.5	78.3	67.5	75.5	9.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	109.5	75.7	66.3	102.2	74.0	64.4	95.8	71.9	36.0	80.6	34.4	78.3	32.7	75.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	16.1	14.3	-35.9	87.7	5.7	3.5	-40.0	90.2	-43.4	92.2	-46.6	94.2	-50.7	96.7		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	5.3	8.4	20.0	35.4	50.0	60.2	63.7	61.6	54.0	41.0	27.6	12.5		
	DBStd	23.06	14.03	12.46	13.52	9.83	9.26	7.44	5.97	6.19	8.34	9.04	10.79	12.56		
	HDD50	6250	1384	1164	932	446	116	10	0	1	53	309	673	1162		
	HDD65	10441	1849	1584	1395	888	472	178	97	144	342	743	1122	1627		
	CDD50	1433	0	0	2	8	118	316	425	359	172	32	1	0		
	CDD65	147	0	0	0	0	8	33	57	38	10	1	0	0		
	CDH74	1416	0	0	0	5	130	369	510	312	86	4	0	0		
	CDH80	285	0	0	0	0	32	77	102	59	15	0	0	0		
	WSAvg	6.7	5.9	6.7	6.9	7.5	7.7	6.9	6.5	6.2	6.5	6.5	6.5	6.0		
	PrecAvg	34.4	2.0	1.6	1.9	2.6	3.1	3.7	3.4	3.5	4.2	3.4	3.1	2.1		
	PrecMax	46.2	3.8	3.7	3.1	5.0	5.9	6.3	6.5	6.1	7.5	6.3	5.7	4.5		
(15) Precipitation	PrecMin	24.9	0.6	0.5	0.1	0.9	1.5	1.5	1.3	1.9	2.1	1.3	1.5	0.6		
	PrecStd	5.4	0.8	0.8	0.7	1.1	1.1	1.2	1.4	1.2	1.3	1.3	1.2	0.9		
	DB	37.4	42.0	61.1	71.8	84.1	86.0	86.4	85.6	82.1	73.0	57.1	40.8			
	MCWB	35.7	37.9	48.9	53.6	67.3	68.6	70.4	70.0	67.3	61.2	50.2	40.3			
	2% DB	33.9	36.0	51.5	64.0	77.0	82.2	82.6	80.7	75.4	66.1	50.1	35.9			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	33.0	33.5	42.4	49.1	60.1	66.1	68.3	67.5	64.4	56.7	46.3	34.5			
	DB	30.4	32.0	44.6	57.5	71.7	78.5	79.2	77.1	71.6	60.6	45.4	33.6			
	MCWB	29.4	29.9	37.7	45.0	55.8	63.3	66.1	65.3	62.3	54.1	42.0	32.6			
	DB	26.7	28.1	39.5	51.9	66.9	74.9	76.7	73.7	67.9	55.3	41.4	31.5			
	MCWB	25.5	26.2	34.2	41.7	53.4	61.4	64.5	63.3	60.6	50.9	39.1	30.8			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	36.7	38.9	58.1	58.1	70.2	71.6	74.0	73.2	70.3	63.0	52.3	40.5			
	MCDB	37.0	41.9	65.1	69.9	81.1	82.0	81.8	80.9	77.8	70.3	57.2	40.6			
	2% WB	33.5	34.4	45.4	52.4	64.4	69.7	71.7	70.5	67.3	59.7	48.0	35.3			
	MCDB	34.0	36.4	51.0	62.0	75.7	79.1	80.8	78.0	74.7	65.6	51.3	36.0			
	5% WB	29.7	31.0	40.7	48.1	60.1	67.4	69.7	68.1	65.1	56.0	43.0	32.7			
(33) Mean Daily Temperature Range	MCDB	30.9	33.0	45.9	58.3	69.5	76.2	79.1	76.0	71.5	61.5	45.5	33.3			
	10% WB	25.9	26.8	36.2	44.6	56.6	65.1	67.4	66.1	62.4	52.0	39.6	30.6			
	MCDB	26.9	28.4	41.1	53.2	66.4	73.3	75.2	73.6	68.9	57.0	42.1	31.5			
	5% DB	19.8	21.9	23.2	21.4	22.9	21.2	20.0	20.0	18.8	15.2	12.6	16.2			
	MCDBR	15.0	20.1	25.6	31.0	30.0	26.9	24.4	23.9	23.8	23.3	17.5	14.3			
(36) Clear-Sky Solar Irradiance	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	taub	0.260	0.268	0.295	0.334	0.352	0.367	0.382	0.362	0.356	0.319	0.284	0.267			
(40) All-Sky Solar Radiation	taud	2.296	2.273	2.311	2.370	2.426	2.396	2.365	2.422	2.439	2.527	2.511	2.317			
	Ebn at Noon	261	285	294	292	289	284	278	279	270	265	255	242			
	Edh at Noon	23	30	34	36	35	36	37	34	30	23	19	20			
(44) All-Sky Solar Radiation	RadAvg	399	736	1200	1508	1694	1804	1769	1477	1107	655	348	277			
	RadStd	70	100	95	133	131	123	134	106	86	84	36	31			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page